Advanced high school · extension

Bézout and integer equations

Progressive exercises

Exercise 1

Find gcd(84,30) and a Bézout identity.

Exercise 2

Does $12x + 18y = 5$ have integer solutions?

Exercise 3

Find all integer solutions of $5x + 3y = 1$.

Exercise 4

Among solutions of $5x + 3y = 31$, find those with nonnegative x and y.